

Mehedi Hasan Bijoy

Speech and Language Processing Researcher

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DEGREES

Doctor of Philosophy (Ph.D.) in Speech & Language Processing (*ongoing*)

- Period : Sep 2025 – Aug 2029 (*planned*)
- Educational Institution : Aalto University, Finland
- Research Topic : Accented Automatic Speech Recognition
- Research Areas : Speech Foundation Models; Large Language Models

Master of Science (M.Sc.) in Computer, Communication, and Information Sciences

- Period : Oct 2023 – Jun 2025
- Educational Institution : Aalto University, Finland
- Major : Speech and Language Technology (*Thesis*)
- Minor : Machine Learning, Data Science, and Artificial Intelligence
- Grade : CGPA 4.83 / 5.00
- Awards/Scholarships : Master's Degree with Honours; Finland Scholarship A (100%) + Grant

Bachelor of Science (B.Sc.) in Computer Science & Engineering

- Period : May 2017 – Sep 2021
- Educational Institution : North South University, Bangladesh
- Specialization : Artificial Intelligence
- Grade : CGPA 3.81 / 4.00
- Awards/Scholarships : Summa Cum Laude; Merit-Based Scholarship

OTHER EXPERTISE

Research Expertise

- Accented/Dialectal Speech Recognition; Low/No-Resourced ASR
- Machine Translation; Sequence-to-Sequence Modeling; Multimodal Representation Learning

Technical Expertise

- Fine-Tuning; Low-Rank Adaptation
- Model Compression; Knowledge Distillation
- Retrieval-Augmented Generation; Prompt Engineering
- Designing, implementing, and deploying end-to-end ML pipelines

LANGUAGE SKILLS

Native Language: Bangla

Other Languages and Proficiency: English - C1/Proficient; Swedish - A1/Beginner (*Learning*)

CURRENT EMPLOYMENT

Sep 2025 – present **Doctoral Researcher**, AaltoASR Research Group, Aalto University, Finland

- Research Focus: Speech and language processing; Accented ASR
- Activity/Outputs: Poster presentation at FCAI Day 2025

PREVIOUS WORK EXPERIENCE

- Mar 2024 – Aug 2025 **Research Assistant**, AaltoASR Research Group, Aalto University, Finland
- Outputs: Publications at *ACMM/MuSe 2024* and *Interspeech 2025*
- Jan 2023 – Oct 2023 **Lecturer**, Department of CSE, Bangladesh University of Business and Technology
- Taught 5 courses: Pattern Recognition, Data Structures, Object Oriented Programming, Structured Programming Language, Discrete Mathematics
 - Teaching scale: ~400 undergraduate Computer Science major students
- Mar 2022 – Jun 2023 **Research Assistant**, IAR, United International University, Bangladesh
- Outputs: Publications at *EMNLP/BLP 2023*, *Computer Speech & Language*, and *Neural Computing and Applications*
- Feb 2022 – Jan 2023 **Lab Instructor**, Department of ECE, North South University, Bangladesh
- Conducted programming labs in C and C++
 - Teaching scale: ~300 undergraduate Computer Science major students

TEACHING MERITS

Teaching Experience

- **Lecturer** (Jan 2023 – Oct 2023), Bangladesh University of Business and Technology: taught 5 CS courses
 - CSE103 - Discrete Mathematics
 - CSE111 - Structured Programming Language
 - CSE121 & 122 - Object Oriented Programming Language and Lab
 - CSE231 & 232 - Data Structures and Lab
 - CSE467 - Pattern Recognition
- **Lab Instructor** (Feb 2022 – Jan 2023), North South University: taught 2 programming lab courses
 - CSE115L - Programming Language I Lab (C)
 - CSE225L - Data Structures and Algorithms Lab (C++)
- **Teaching Assistant**, Aalto University
 - ELEC-E5510 – Speech Recognition D (Autumn 2024, Autumn 2025)
 - ELEC-E5550 – Statistical Natural Language Processing D (Spring 2025, Spring 2026)
- **Teaching Assistant**, North South University
 - MAT361 – Probability & Statistics (Fall 2020, Spring 2021, Summer 2021)

AWARDS & HONOURS

- **High Quality Teaching Award** for ELEC-E5510 Speech Recognition D (2024–2025), Aalto University
- **Honours** for academic result in MSc, Aalto University
- **Summa Cum Laude** for academic result in BSc, North South University

RESEARCH FUNDING & GRANTS

- **SaameAI Grant** (to support PhD studies), Aalto University
- **Finland Scholarship Grant** (to support MSc studies), Aalto University
- **Merit-Based Scholarship** (to support BSc studies), North South University

OTHER KEY ACADEMIC MERITS

Reviewer

- Workshop on Bangla Language Processing at IJCNLP-AAACL 2025
- Workshop on Bangla Language Processing at EMNLP 2023

RESEARCH OUTPUT

Overview of Outputs

- Peer-reviewed publications: 7
- Poster: 1
- Presentations: MSc thesis presentation at Aalto University; Paper presentation at INTERSPEECH 2025

Key Research Contributions

- Proposed a Self-Supervised Accent-Discovery method with Fairness-Aware Adapter Routing that outperformed zero-shot, fine-tuning, and LoRA baselines while using only ~0.7% of FT parameters.
- Improved accented English ASR robustness by ~14% relative WER using a novel Multi-Teacher Knowledge Distillation framework evaluated across 30+ accents from L1 and L2 speakers.
- Achieved 5–12% performance gains in multilingual speech emotion recognition for English, Finnish, and French by leveraging Language-Aware Mixture-of-Experts architecture.
- Developed two large-scale Bangla corpora for grammatical error correction and punctuation restoration, yielding >10% performance gains on both tasks.

Method, Software, Infrastructures, and Tools Developed

- Whisper-powered Speech-to-Text web application (FastAPI, PyTorch, HuggingFace Transformers)
- RAG-based question-answering system (Llama, FAISS, LangChain)

Posters

- At FCAI Day 2025

Publications

- **Conference** **Mehedi Hasan Bijoy**, Dejan Porjazovski, Tamás Grósz, and Mikko Kurimo. "Multi-Teacher Language-Aware Knowledge Distillation for Multilingual Speech Emotion Recognition." Proc. Interspeech 2025, 146-150, doi: 10.21437/Interspeech.2025-418
- **Journal** **Mehedi Hasan Bijoy**, Nahid Hossain, Salekul Islam, and Swakkhar Shatabda. "A transformer-based spelling error correction framework for Bangla and resource scarce Indic languages." Computer Speech & Language 89 (2025): 101703.
- **Workshop** **Mehedi Hasan Bijoy**, Dejan Porjazovski, Nhan Phan, Guangpu Huang, Tamás Grósz, and Mikko Kurimo. "Multimodal Humor Detection and Social Perception Prediction." In Proceedings of the 5th on Multimodal Sentiment Analysis Challenge and Workshop: Social Perception and Humor, pp. 60-64. 2024.
- **Journal** Nahid Hossain*, **Mehedi Hasan Bijoy***, Salekul Islam, and Swakkhar Shatabda. "Panini: a transformer-based grammatical error correction method for Bangla." Neural Computing and Applications 36, no. 7 (2024): 3463-3477.
- **Journal** **Mehedi Hasan Bijoy**, Nirob Hasan, Mithun Biswas, Suvodeep Mazumdar, Andrea Jimenez, Faisal Ahmed, Mirza Rasheduzzaman, and Sifat Momen. "Towards sustainable agriculture: a novel approach for rice leaf disease detection using dCNN and enhanced dataset." IEEE Access 12 (2024): 34174-34191.
- **Workshop** **Mehedi Hasan Bijoy**, Mir Fatema Afroz Faria, Mahbub E. Sobhani, Tanzid Ferdoush, and Swakkhar Shatabda. "Advancing bangla punctuation restoration by a monolingual transformer-based method and a large-scale corpus." In Proceedings of the First Workshop on Bangla Language Processing (BLP-2023), pp. 18-25. 2023.
- **Conference** **Mehedi Hasan Bijoy**, Nirob Hasan, Md Tahrim Faroque Tushar, and Shafin Rahman. "Image tagging by fine-tuning class semantics using text data from web scraping." In 2021 24th International Conference on Computer and Information Technology (ICCIT), pp. 1-6. IEEE, 2021.

OTHER MERITS

- **Programming Languages:** Python, C, C++, Java
- **ML Libraries & Frameworks:** PyTorch, HuggingFace Transformers, LangChain, Ollama
- **MLOps:** MLflow, Docker, Airflow, GitHub Actions, AWS, Flask, FastAPI, Grafana
- **Data Storage & Acquisition:** SQL Databases, Vector Databases, Web Scraping (Selenium, BeautifulSoup)
- **Developer Tools:** Git, DVC, Linux, LaTeX